Capstone release review

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Grounded Learning Service

WORKBOOK · 1 OF 15

Recall and choose your starting point

> Record your starting explanation for Capstone release review before opening notes. Show any later correction separately.

Try the recall before opening notes. Then use a different colour or a labelled second line to correct your explanation.

Capstone failure testing

Without opening the old lesson, explain one key idea from Capstone failure testing. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Documentation and operational readiness

Without opening the old lesson, explain one key idea from Documentation and operational readiness. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Performance and cost experiments

Without opening the old lesson, explain one key idea from Performance and cost experiments. Give an example and a mistake that the idea helps you avoid.

My remembered idea, example and correction

Lesson reference for workbook practice

Use this reference for the original W1–W3 and W5 exercises. The separately named independent case supplies its own changed facts; use those facts for that attempt before feedback.

Release review combines correctness, access, failure recovery and operational readiness. **Critical gate**s can block release independently of an average score.

Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. Record each gate separately with versioned **evidence**.

Practise with fading help

> Complete the supported Capstone release review trace and record the first step you can explain without help.

Use W2: Trace the worked case for Capstone **release review**. Record the answer once; the lesson and workbook refer to the same attempt.

Given inputs	Procedure I chose	Middle steps	Result and check

> **The step I completed without help, and the hint I needed**

Individual reasoning check

> Check the proposed Capstone release review explanation against the supplied facts before discussing it.

Use the worked case for Capstone **release review**. Proposed explanation: “The worked result establishes every future case”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

My explanation before partner discussion

WORKBOOK · 4 OF 15

Independent transfer case

> Write one first attempt at the reserved independent case before feedback. Repeated copies of that case refer to the same recorded attempt.

Independent case: Apply a new case independently

Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

My independent explanation, trace and limits

Attempt this case before feedback. Record any help. Earlier examples below are further practice; a case already practised with feedback cannot establish fresh transfer.

WORKBOOK · 5 OF 15

W1 Explain the mechanism

> Explain the mechanism. Apply the supplied Capstone release review facts and keep your

> reasoning.

Explain the weekly **mechanism** in your own words. Use the worked scenario to name its **input**, decision rule and **output**.

My reasoning and evidence

WORKBOOK · 6 OF 15

W2 Trace the worked case

> Trace the worked case. Apply the supplied Capstone release review facts and keep your reasoning.

Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. Show the intermediate steps and give the conclusion with its scope.

My reasoning and evidence

WORKBOOK · 7 OF 15

W3 Audit the limit

> Audit the limit. Apply the supplied Capstone release review facts and keep your reasoning.

A report omits this qualification: Release readiness is scoped to the declared gates and environment; deployment still needs operational checks.

Explain how omitting it could change a reader's interpretation, and write an accurate replacement.

My reasoning and evidence

W4 Apply a new case independently

> Attempt before feedback. If this repeats the selected independent case, reuse that first attempt; otherwise keep a separate record. Apply a new case independently. Apply the supplied Capstone release review facts and keep your reasoning.

Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

My reasoning and evidence

WORKBOOK · 9 OF 15

W5 Design a revealing follow-up

> Design a revealing follow-up. Apply the supplied Capstone release review facts and keep your reasoning.

For the worked scenario, propose one additional case that could reveal a failure, describe its expected outcome before running it, and explain what would need to remain fixed for a fair comparison.

Do not reuse W4 as a new final test after receiving feedback.

My reasoning and evidence

WORKBOOK · 10 OF 15

Project checkpoint and reflection

> Connect Capstone release review to Grounded Learning Service, identify a limit and choose the next useful test.

For Grounded Learning Service, use this week's independent task as a rehearsal for the final **evidence** pack: Ordinary 27/30 pass; access 1/5 fails; rollback passes; no effects. Decide.

Keep actual **observation**s separate from expected results, preserve failures, compare a baseline and explain the limits.

My project change, supporting evidence and remaining limit

WORKBOOK · 11 OF 15

Feedback, return check and learning record

> Record the first repaired step and a different case to revisit after a delay.

Evidence	My record
Worked alone / hint / model	
First step I repaired and why	
New case checked later; date and result	
What I can now explain without notes	
One remaining question	

Revisit this idea with your teacher after a delay. Familiarity is different from explaining and using it on a changed case.

WORKBOOK · 12 OF 15

Pictorial case practice: Capstone release review

> Review every mandatory gate. A capstone release needs factual, access and deletion gates

> to pass. Explain the labelled step before checking the changed condition.

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.



Follow the labelled picture with the matching explanation. Open the vector diagram

> Apply the three mandatory release gates to the recorded results. Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

My picture-based explanation and changed-case prediction

Record the help used

Support used: alone / hint / worked example. This record is practice **evidence**.

Project connection: For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

WORKBOOK · 13 OF 15

Project portfolio milestone

> Keep this week's input, deciding step, result and limit together in the Grounded Learning Service record.

Project: Grounded Learning Service

Current milestone: Present and defend. Demonstrate the system, explain one failure and justify the next improvement with **evidence**.

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Evidence to keep	My record
This week's input and deciding step	
Prediction and actual result (or labelled paper trace)	
One change since the previous checkpoint and why	
One error or limit and the next test	

Explain what this week adds to your project. Reuse last week's record rather than starting again.

WORKBOOK · 14 OF 15

Supported practice and learning evidence

> Try the supported questions before comparing with hints and feedback. Record what you repaired and the evidence you retained.

Use this supported practice once, in the browser or on paper. Keep the reserved independent assessment for a separate first attempt before teacher feedback.

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass.

Practice 1: explain the deciding evidence

Which labelled step matches this detail? Remove the listed copy and verify it before a fresh review.

1. List **evidence**

2. Specify repair

3. Apply decision rule

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Read the steps in the pictorial case. Match the supplied detail to the stage that performs or records it.

Feedback · compare after trying

After your attempt, compare with the supported explanation: Specify repair Remove the listed copy and verify it before a fresh review. **Release review** applies explicit conditions to a specific candidate version.

Completed coding is not **evidence** that all operational obligations are satisfied.

Practice 2: explain the deciding evidence

Changed case · predict first

Changed-case practice: Every listed deletion location passes new verification. Which conclusion is supported by the supplied facts?

1. This one example guarantees correct results for every future **input**.

2. We can decide the result without examining the changed **input** or the procedure.

3. That gate can pass; review the complete current **evidence** before approving release.

My selected conclusion, deciding fact and explanation

Hint · try first

Hint if needed: Use the changed condition and the deciding step in the pictorial trace. The answer must say what follows from those facts.

Feedback · compare after trying

After your attempt, compare with the supported explanation: That gate can pass; review the complete current **evidence** before approving release. One repaired gate does not automatically refresh **evidence** for every other gate.

Repair a missing building block

Review Capstone failure testing (ai-6/34). Goal: Explain the **mechanism** and terminology of capstone failure testing. Short practice: Explain why retrieval success permits the checked 15:00 fallback when drafting fails.

What changes if retrieval fails too, and why can a template not fill that **evidence** gap?. Return to this week and explain what you changed.

Review Documentation and operational readiness (ai-6/32). Goal: Explain the **mechanism** and terminology of documentation and operational readiness. Short practice: Match SOURCE_UNAVAILABLE to its safe check and escalation owner.

What is lost when the maintainer route is removed, even if the diagnostic instructions stay readable?. Return to this week and explain what you changed.

Review Performance and cost experiments (ai-6/29). Goal: Explain the **mechanism** and terminology of performance and cost experiments. Short practice: Calculate per-case seconds and cost units for A and B.

Which meets a budget of at most 2 cost units per case, and what speed trade-off does that choice make?. Return to this week and explain what you changed.

Keep your completed work

Use Download completed work in the hosted lesson to export your answers, reflections, practice feedback and code-run **evidence**. Open the HTML copy to read or print it.

Drafts stay in this browser when storage is available; clear drafts on a shared device.

What I repaired, the evidence I kept and the next test

WORKBOOK · 15 OF 15

Visual concept journey

> Explain each Capstone release review picture from its labels, then record a relevant limit.

01. Capstone release review: the case



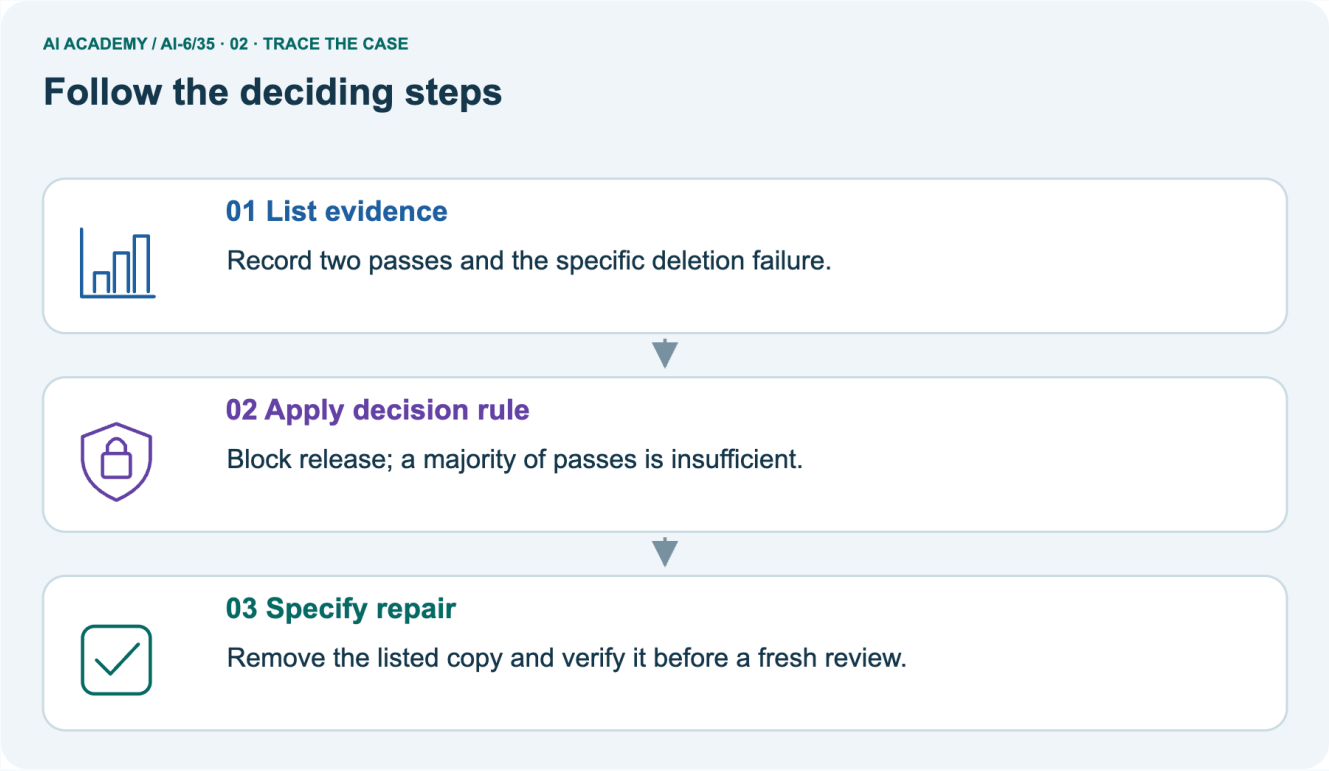
Follow the labelled picture with the matching explanation. Open the vector diagram

Review every mandatory gate. A capstone release needs factual, access and deletion gates to pass. Supplied facts: Factual and access checks pass. Deletion fails because a listed backup remains. All three gates are mandatory.

> Point to the given fact that matters most. Explain why it matters.

My picture explanation, deciding evidence and one limit

02. Follow the deciding steps



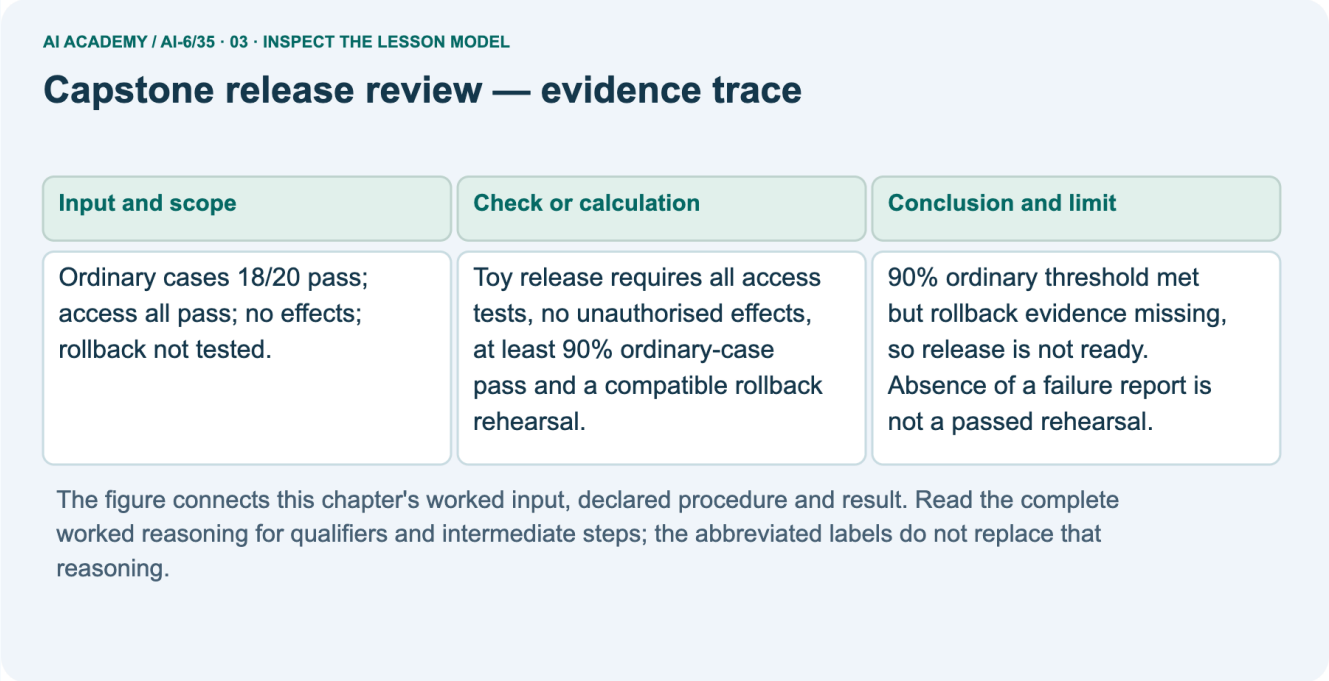
Follow the labelled picture with the matching explanation. Open the vector diagram

List **evidence**: Record two passes and the specific deletion failure. → Apply decision rule: Block release; a majority of passes is insufficient. → Specify repair: Remove the listed copy and verify it before a fresh review.

> Point to each numbered stage and explain which recorded **input** it uses.

My picture explanation, deciding evidence and one limit

03. Capstone release review — evidence trace



Follow the labelled picture with the matching explanation. Open the vector diagram

Capstone **release review** — **evidence** trace.

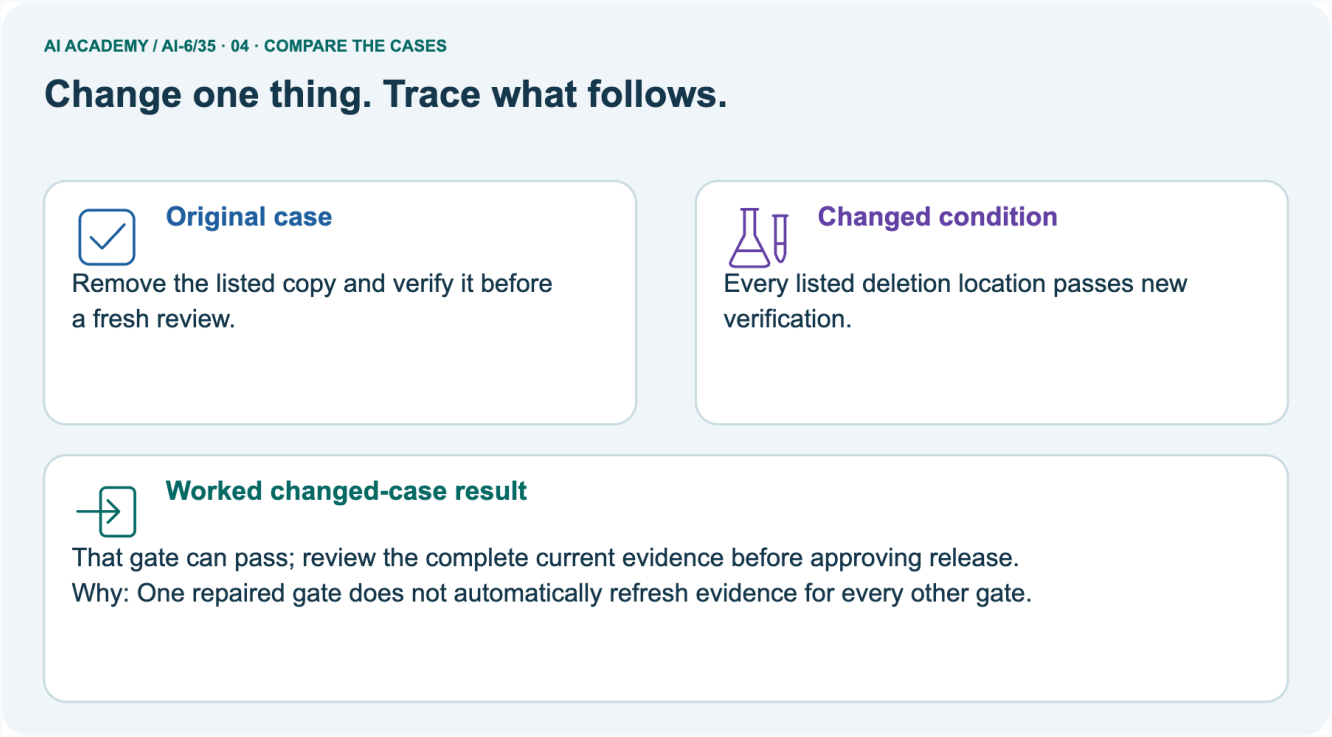
Ordinary cases 18/20 pass; access all pass; no effects; rollback not tested. | Toy release requires all access tests, no unauthorised effects, at least 90% ordinary-case pass and a compatible **rollback rehearsal**. | 90% ordinary threshold met but rollback **evidence** missing, so release is not ready.

Absence of a failure report is not a passed rehearsal.

> Read the labels and values. Explain the relationship the model shows.

My picture explanation, deciding evidence and one limit

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

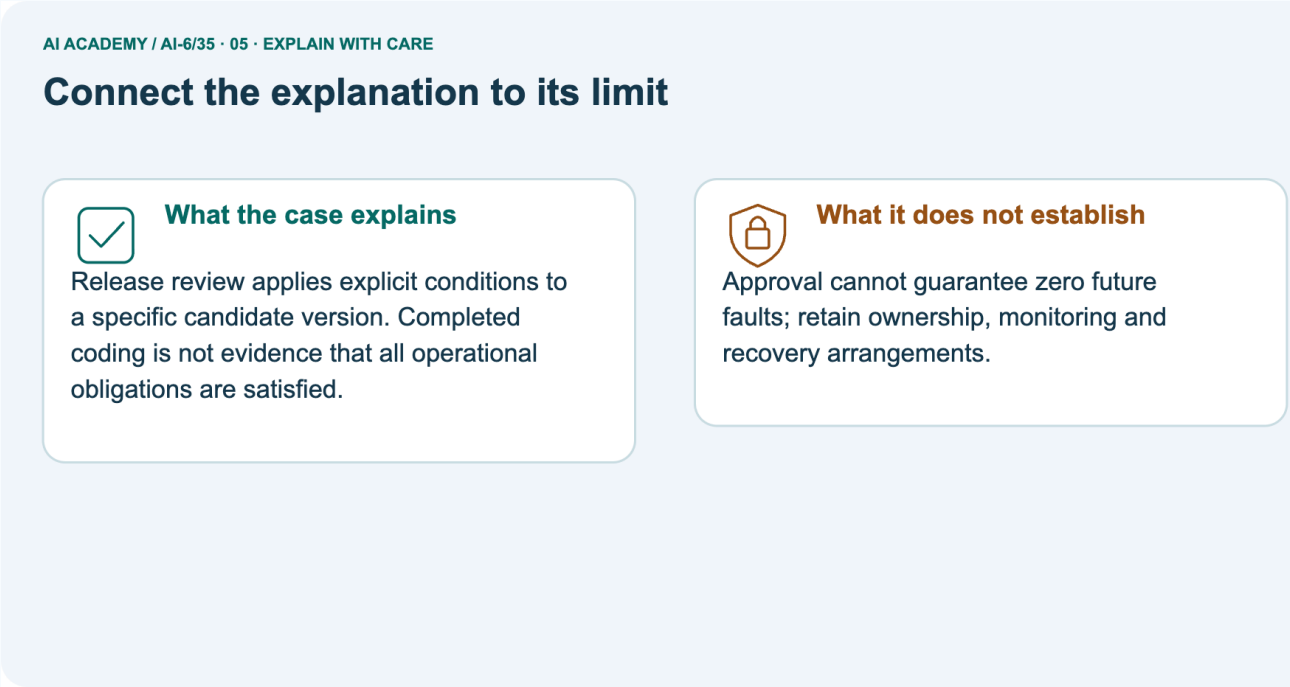
Original case: Remove the listed copy and verify it before a fresh review. Changed condition: Every listed deletion location passes new verification.

Expected result: That gate can pass; review the complete current **evidence** before approving release. Reason: One repaired gate does not automatically refresh **evidence** for every other gate.

> Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

My picture explanation, deciding evidence and one limit

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

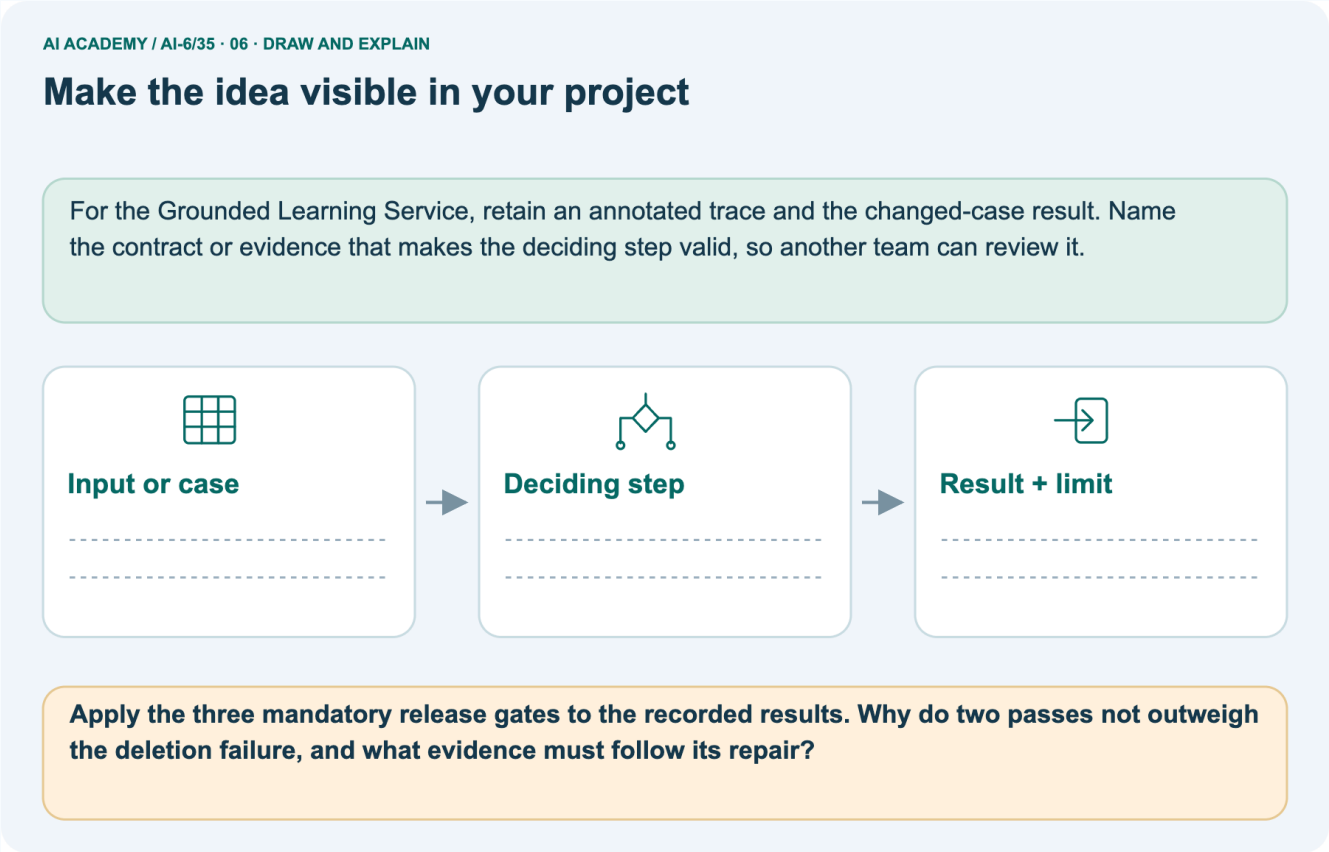
Release review applies explicit conditions to a specific candidate version. Completed coding is not **evidence** that all operational obligations are satisfied.

Limit: Approval cannot guarantee zero future faults; retain ownership, monitoring and recovery arrangements.

> Explain the case, then point to a limit on the claim.

My picture explanation, deciding evidence and one limit

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

For the Grounded Learning Service, retain an annotated trace and the changed-case result. Name the contract or **evidence** that makes the deciding step valid, so another team can review it.

Draw your **input**, deciding step, result and limit. Supported question: Apply the three mandatory release gates to the recorded results.

Why do two passes not outweigh the deletion failure, and what **evidence** must follow its repair?

> Draw your project trace and justify the deciding step using **evidence**.

My picture explanation, deciding evidence and one limit

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.